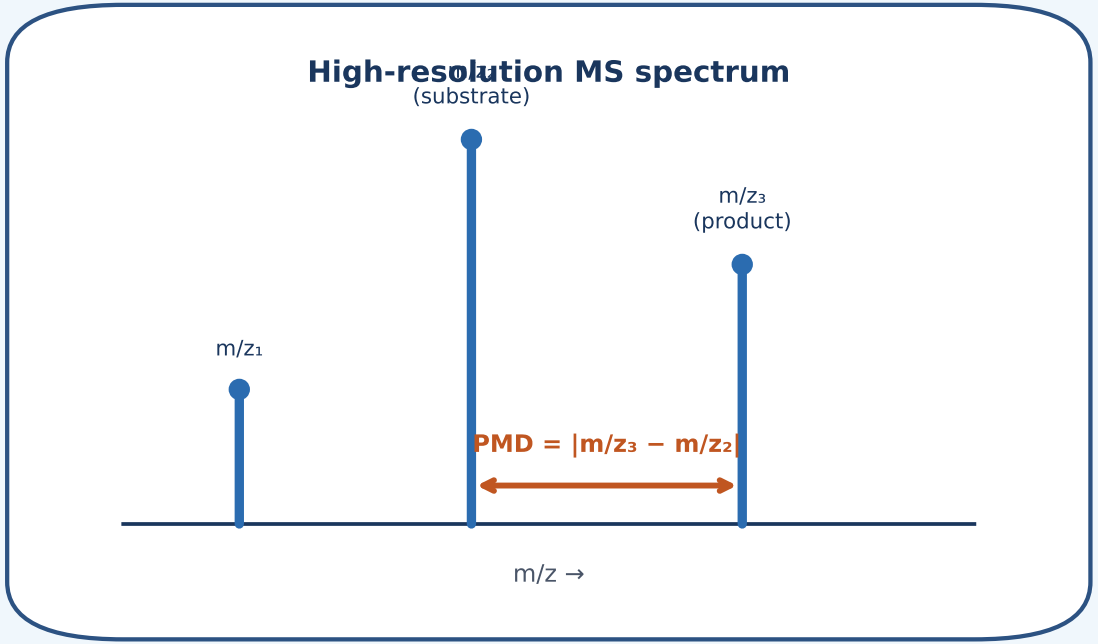
Mass Spectrometry as a Chemical Reaction Detector

A fixed paired mass distance (PMD) between two ions encodes a specific biotransformation



PMD → Reaction

2.0157 Da	H_2	Reduction
14.0157 Da	CH_2	Methylation
15.9949 Da	O	Oxidation
18.0106 Da	H_2O	Hydration
79.9663 Da	HPO_3	Phosphorylation
162.0528 Da	$C_6H_{10}O_5$	Glycosylation

Core insight: fixed Δm between ions = chemical reaction linking them

Requires ~5 ppm accuracy (Orbitrap / FT-ICR / Q-TOF)